

Econ 2 - Lecture 12 - 5/11/20

Lecture Quiz #6 Released Wednesday, due Monday, 5/18

Discussion Section this week $\rightarrow$ Traditional Section

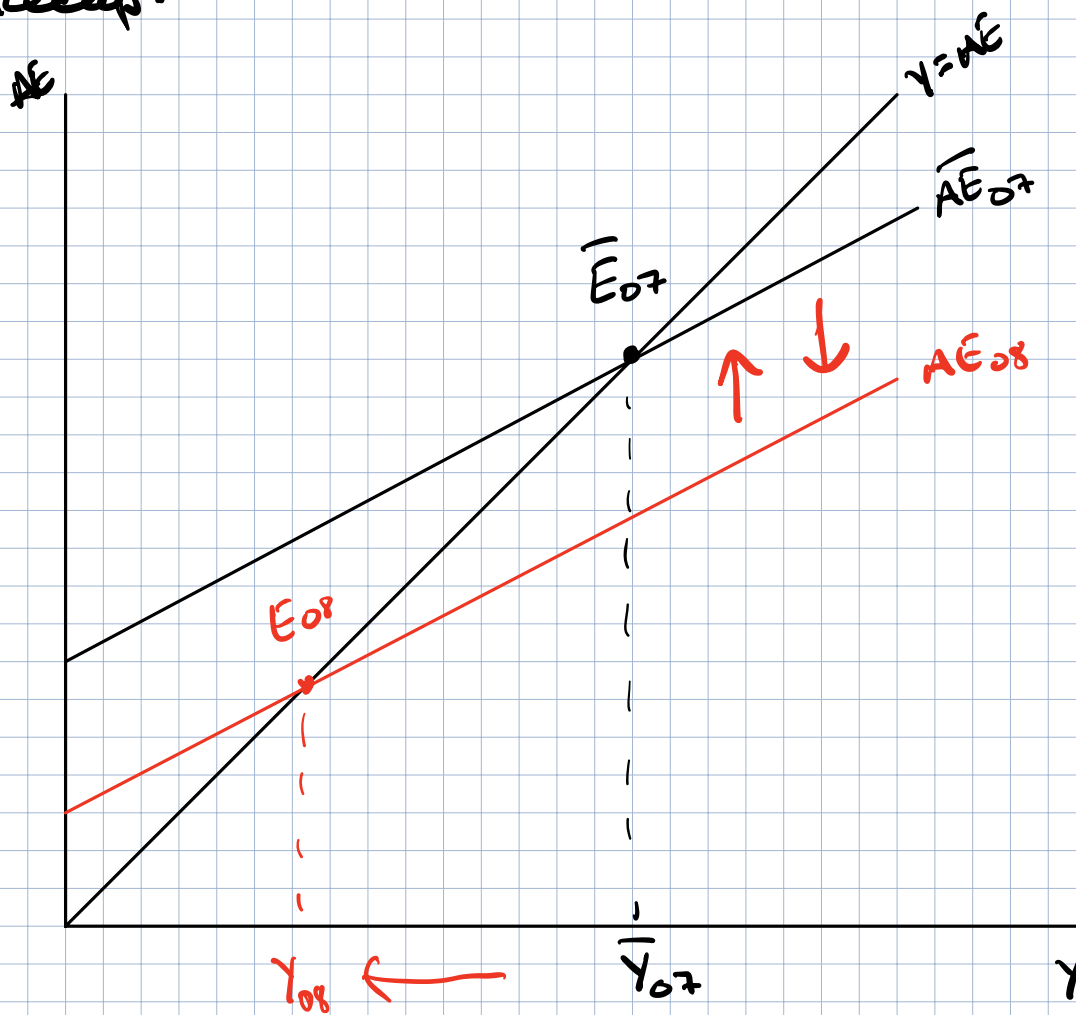
Next Week: Discussion Activity #4 $\rightarrow$ Gamification $\rightarrow$ Passcode
Coming

Last Class: Using $Y = AE$ to explain macroeconomy

$\rightarrow$ Multiplier Effect $\Rightarrow$ Expenditure Multiplier

$\rightarrow$ Policy Actions? Ask for help to increase AE?

Recap:



Start in 2007
 $\rightarrow \bar{Y}$, full employment

Housing Crash

$\rightarrow \downarrow AC, \downarrow I^p$

$\rightarrow$ Drop $\bar{Y}$ to Y_{08}

$\rightarrow$ Cyclical UE

$\rightarrow$ Asked for
help $\rightarrow$
 $\Delta I^p > 0$

Instead of Policy makers asking for $\Delta I^p > 0$

Job of Federal Government is to bring Y back to $\bar{Y}$

What can Government (President, Congress)
change in AE model?

Change $G \rightarrow$ Build a new highway

Change $T \rightarrow$ Tax cut

Fiscal Policy

1. Δ in G

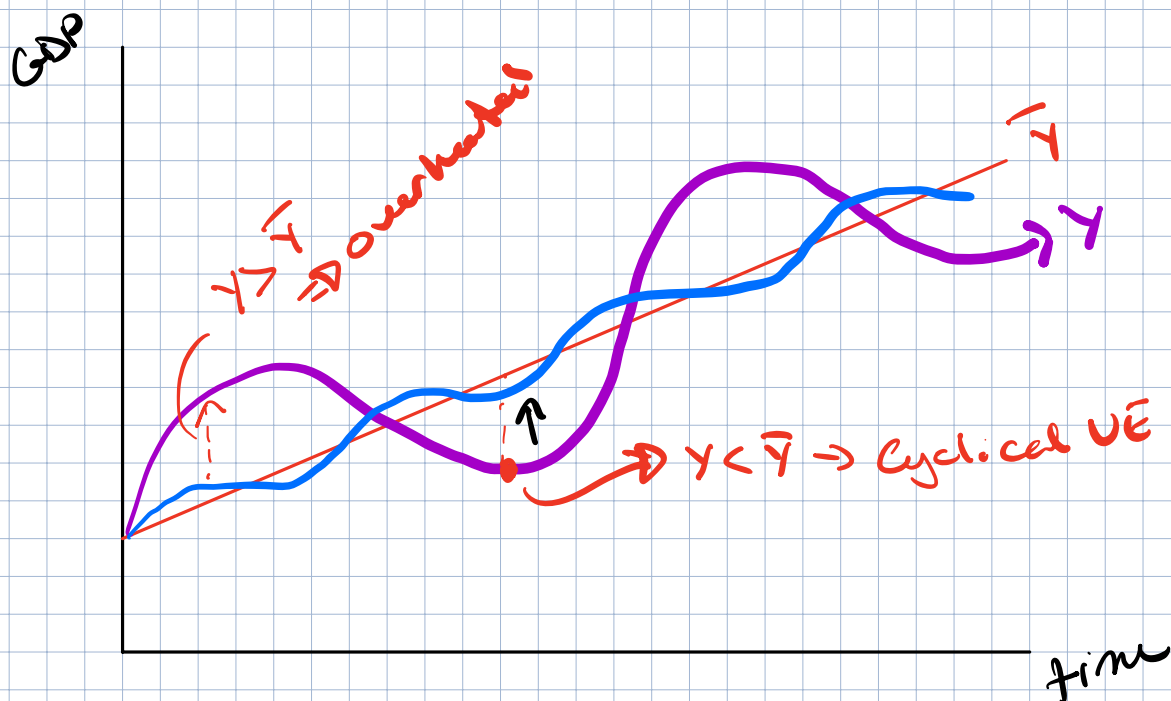
2. Δ in T

If $Y_{08} < \bar{Y} \rightarrow$ what direction?

$\Delta G > 0 \Rightarrow \uparrow AE$

$\Delta T < 0 \Rightarrow \uparrow AE$

Big Picture Goal $\rightarrow$ Business Cycle



Gov't will want to "counter" business cycle

Gov't will carryout counter-cyclical fiscal Policy

Define Fiscal Policy Categories

1. Discretionary Fiscal Policy $\rightarrow$ Choice

$\Delta G \Rightarrow$ Choice about the direction & magnitude

$\rightarrow$ Defense spending, infrastructure, NASA, TSA, National Parks

$\Delta T \Rightarrow$ Individual taxes (gasoline), tax brackets

2. Automatic Stabilizers

Changes in G or T that occur automatically

As Y increases $\rightarrow$ automatic increase in T

As $\Delta T > 0 \Rightarrow \Delta Y < 0 \rightarrow$ stabilizer

As Y decreases $\rightarrow$ increase in Unemployment $\rightarrow$ Unemployment

As $\Delta G > 0 \Rightarrow \Delta Y > 0$ Benefits
 $\Delta G > 0$

Apply Fiscal Policy to Analysis

$Y_1 = 7,500$, $\bar{Y} = 10,000$, $mpe = 0.9$

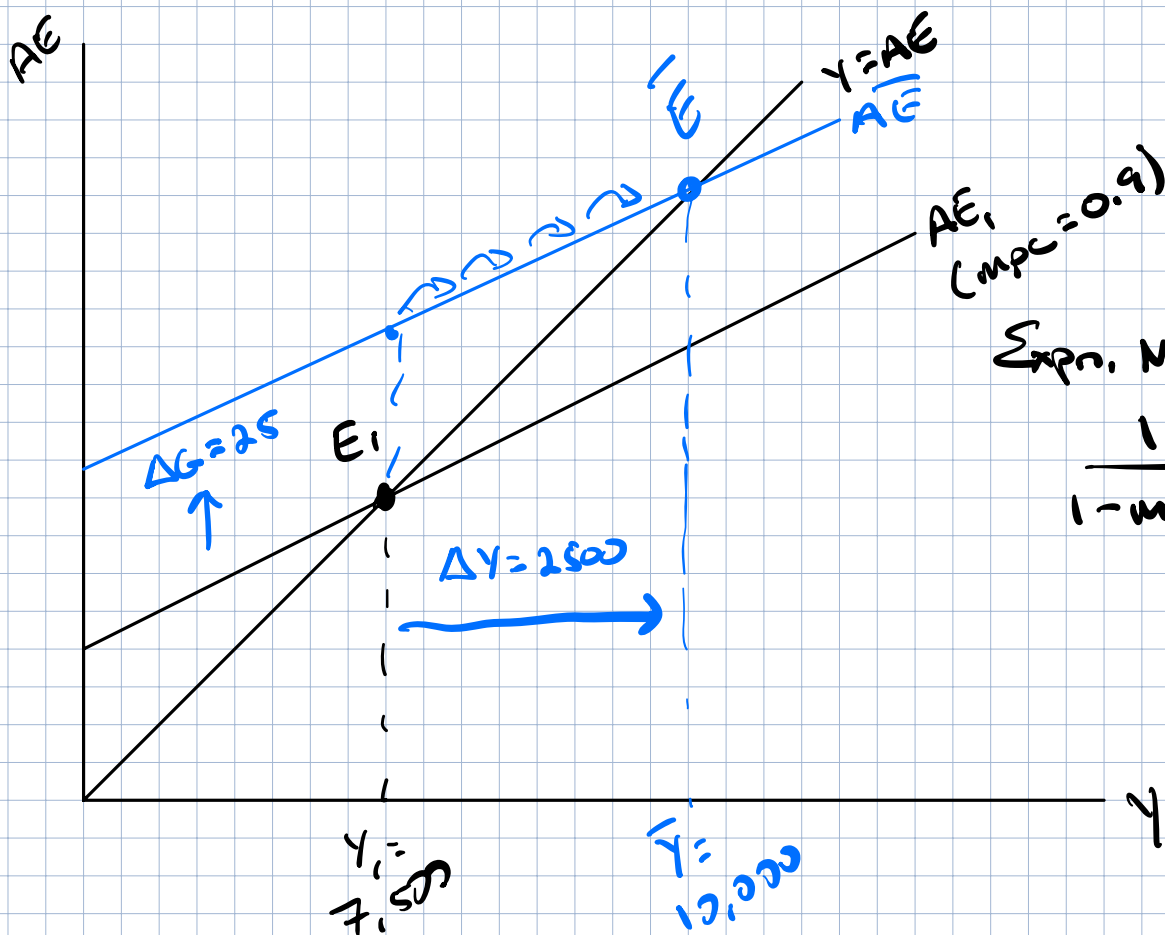
How much should we ΔG in order to get back to $\bar{Y}$?

Draw Graph:

Apply Fiscal Policy to Analysis

$$Y_1 = 7,500, \bar{Y} = 10,000, mpc = 0.9$$

How much should we ΔG in order to get back to $\bar{Y}$?



Expr. Multiplier = 10

$$\frac{1}{1-mpc} = \frac{1}{1-0.9} = \frac{1}{0.1}$$

$$\Delta Y = \frac{1}{1-mpc} \Delta G \rightarrow 2500 = 10 \times \Delta G, \Delta G = 250$$

How does the government actually change G ?

Project that creates jobs

↳ Propose bill in House. Send to Senate. Agree

↳ Pass on to President (no Veto)

Bridge in Cincinnati $\Rightarrow$ I-71/75

$\rightarrow$ Used to be #1 infrastructure problem in US $\times$ #2

$\rightarrow$ Bad Shape + 3-4% of GDP travel across each year

2012: Rally with Obama + Rand Paul want AG
Propose a bill $\rightarrow$ Senate $\rightarrow$ Rejected $\rightarrow$ Mitch McConnell
 $\rightarrow$ Too much in bill

Still slow with agreements

If we decide to build bridge

Who builds? Bids for bridge

$\rightarrow$ Cheap bridge X

$\rightarrow$ Expensive bridge $\rightarrow$ Gold Plates $\rightarrow$ X

$\rightarrow$ Cousin $\rightarrow$ build bridges X

Alternatives to AG $\rightarrow$ ΔT

Instead of $\Delta G = 250$, $\Delta T = -250$

Compensate

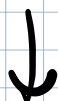
$$\Delta G = 250$$

$$\Delta Y = 250$$



$$\Delta C = 0.9 \times 250$$

$$\Delta Y = 225$$



$$\Delta C = 0.9 \times 225$$

$$\Delta Y = 202.50$$

$\vdots$

$$\text{Total } \Delta Y = 2500$$

$$\Delta Y = 10 \times 250$$

$$\Delta T = -250$$

Tax Cut is not production



$$\Delta C = 0.9 \times 250$$

$$\Delta Y = 225$$



$$\Delta C = 0.9 \times 225$$

$$\Delta Y = 202.50$$

$\vdots$

Round 1



$$\text{Total } \Delta Y = 2500 - 250$$

$$\Delta Y = 2250$$

$$\Delta Y = 9 \times 250$$

Tax Multiplier $\neq$ Expenditure Multiplier
 $\rightarrow$ Fraction of Expen. Multiplier

$$\text{Tax Mult.} = \frac{-\text{MPC}}{1-\text{MPC}} \rightarrow \frac{-0.9}{1-0.9} = -9$$

$$\Delta Y = -9 \times \Delta T = -9 \times -250 = 2250$$

$$\Delta Y = 4000, \text{MPC} = 0.8, \Delta T = ?$$

$$4000 = \frac{-0.8}{1-0.8} \Delta T = \frac{-0.8}{0.2} \Delta T = -4 \cdot \Delta T$$

$\Delta T < 0 \Rightarrow \Delta Y > 0$

$$\frac{4000}{-4} = -1000 = \Delta T$$

$|\text{Tax Multiplier}| < |\text{Expenditure Multiplier}|$

Magnitude of Tax Multiplier is less than Expen

Imagine only one group can receive tax cut $\rightarrow$ which one?

Option #1: Give tax cut to low-income households

Benefit: Spend a high fraction of tax cut immediately

MPC is relatively high $\rightarrow \text{MPC} = 0.95$

$$\text{Tax Multiplier} = -\frac{0.95}{1-0.95} = -19$$

Limitations: $\Delta Y = -\frac{\text{MPC}}{1-\text{MPC}} \Delta T$

Income = 30K, $T = 64\text{K/year} \rightarrow \Delta T = -4000$, ceiling on ΔT

Option #2: Give tax cuts to high-income

Benefit: Business Owners/Investors/Etc. and
invest in production $\rightarrow$ create high paying jobs

ΔT can be very large

* 1 million / yr salary $\Rightarrow$ Federal Taxes = 350K / yr

A small percentage cut $\Rightarrow$ big ΔT

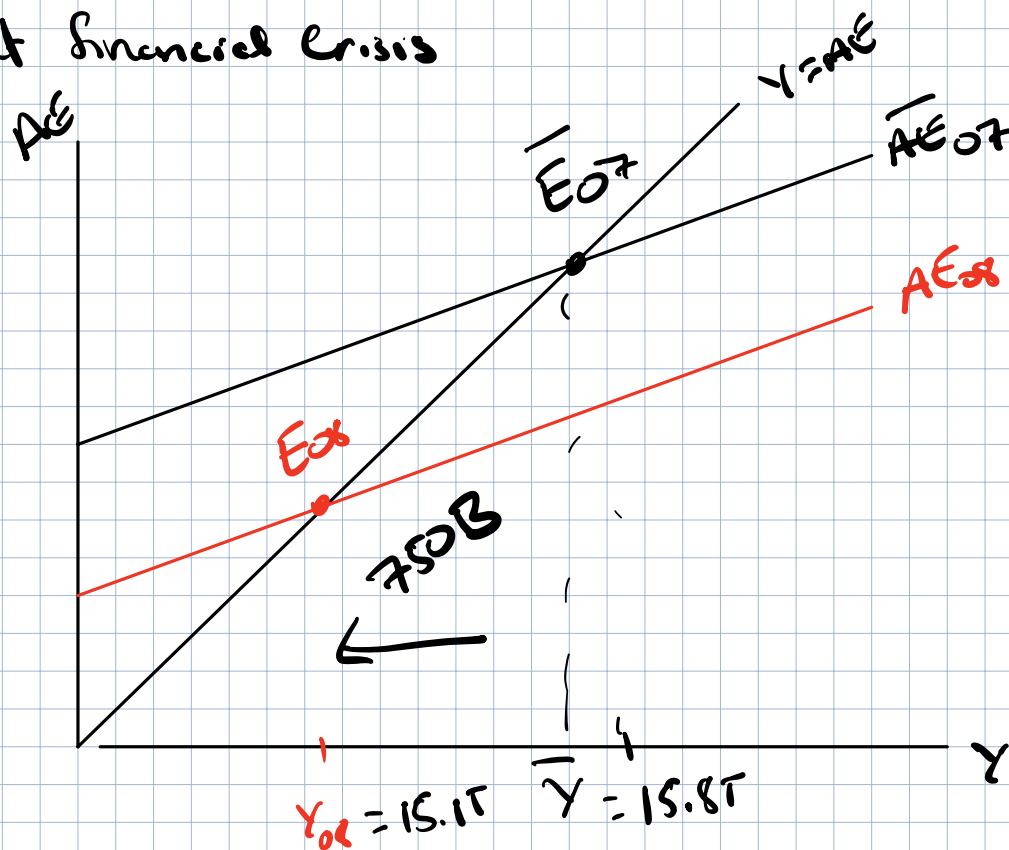
Limitations: MPC is possibly low

$$\text{If } mpc = 0.5, \text{ tax multiplier} = \frac{-0.5}{1-0.5} = -1$$

Possible that some individuals have $mpc = 0$

Fiscal Policy is difficult to change in big way

Great Financial Crisis



Gov't enacted Fiscal Policy

2008: Bailout Bill → Loan given by US Treasury

Goldman Sachs, JP Morgens, Citigroup, Big 3 Auto

→ Allowed to borrow up to ¹⁵700 B

→ Pay a 5% Dividend Annually (up to 9% in 2013)

→ US Treasury made \$15 billion in profit

2009: American Recovery & Reinvestment Act
(ARRA) → Economic Stimulus
Package